Part -2

Advanced NMap

A broad overview and the basic features of NMap have been covered in an earlier article in *LINUX For You*. In this series, we discuss in detail various NMap scan types, and the practical use of these commands to scan various devices and networks.

Before we begin understanding NMap scan types, let us start with the basics, including understanding the 3-way TCP handshake. TCP/IP is not a single protocol, but a suite comprising various protocols, some of which are detailed in Table 1.

Table 1: Various TCP/IP protocols

1.	Application layer	FTP, HTTP, SNMP, BOOTP, DHCP
2.	Transport layer	TCP, UDP, ICMP, IGMP
3.	Network layer	ARP, IP, RARP
4.	Data link layer	SLIP, PPP

UDP and TCP

UDP is a connection-less protocol that does not assure the delivery of packets at the other end. However, that does not mean it is an unreliable protocol; higher-level applications must take care to verify that data has been received at the other end. This practice has its own uses, like with live audio/video transfers, where real-time delivery is a must.

TCP is a connection-oriented protocol, which assures delivery of packets. ICMP packets are used to convey error messages, if any. The TCP three-way handshake is used to establish and reset connections, and this concept is key to understanding various NMap scan types. In the TCP three-way handshake:

1. A 'client' initiates communication with a SYN (Synchronise) packet with a randomly generated number, X.
2. The server acknowledges with a SYN–ACK (Acknowledgement), X+1 and a randomly generated number, Y.
3. The client again sends an ACK, followed by Y+1, thus completing the handshake. Now the client and server can start data transfer.

After the data transfer is complete, a FIN (Finish) packet is sent by the client, to end the connection. NMap uses/tweaks this handshake very effectively for various scan types. Before we proceed, let us be clear about two basic but important aspects of NMap scans:

1. By default, NMap scans 1000 most common ports for each protocol. The list of these ports can be modified in the *nmap-services* file, typically stored in */etc/services*. (I have never used this; the default ports are almost always sufficient!
2. Root privileges are required to run any scan that modifies the standard TCP handshake.
3. Now, let us try to understand the detailed workings of various NMap scan types.

TCP SYN Scan –sS

This is the default NMap scan, used to detect open TCP ports in the target range. At the start of a SYN Scan, NMap initiates a TCP handshake with a standard SYN packet, to the required TCP port of the device to be scanned (target). The target's response, giving details of port status, differs depending on the status of the destination port (see Table 2).